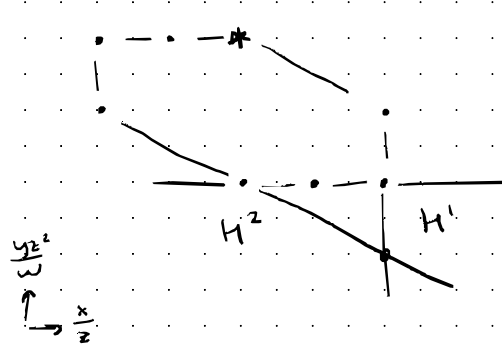
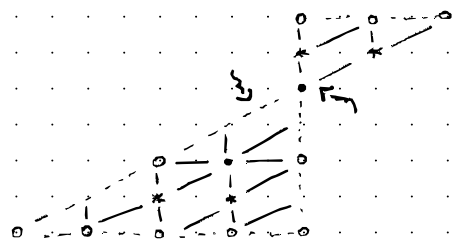
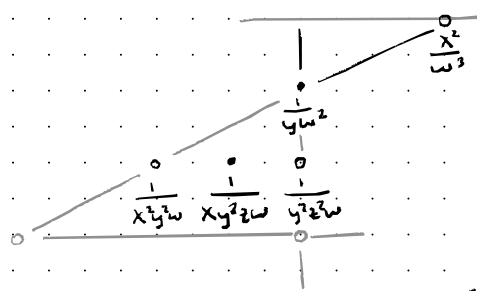
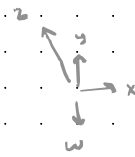


* what does the spectral sequence do to a cosymology
 elt that would map to 0 but instead goes to
 a different type?

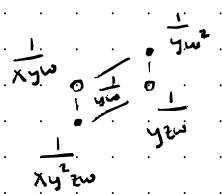


first degree with
 multiple types.

(2, -3)



should represent $H^2(F_2) \rightarrow H^1(F_0)$

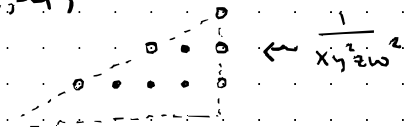


$H^2(\mathcal{O}(0, -1)) \rightarrow H^1(\mathcal{O})$ because
 for $\square$ not *

well before arising by $\mathcal{O}(2, -3)$

$$\mathcal{O}(0, -1) \otimes \mathcal{O}(2, -3) = \mathcal{O}(2, -4)$$

$H^2(F_1) \rightarrow H^2(\mathcal{O}(1, -1)) \rightarrow \mathcal{O}(3, -4)$



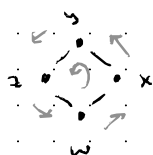
$$\tilde{H}_1 \otimes \begin{cases} \text{still in} \\ \text{homology} \end{cases} \left\{ \begin{bmatrix} -z \otimes \frac{1}{xy^2zw^2} \\ x \otimes \frac{1}{xy^2zw^2} \\ -1 \otimes \frac{1}{xy^2zw^2} \\ 1 \otimes \frac{1}{y^2zw^2} \end{bmatrix} \leftarrow 1 \otimes \frac{1}{xy^2zw^2} \text{ in single summand} \right. \\ \left. \begin{cases} \text{not in} \\ \text{homology} \end{cases} \right\} \left\{ \begin{bmatrix} -z \otimes \frac{1}{xy^2zw^2} \\ x \otimes \frac{1}{xy^2zw^2} \\ -1 \otimes \frac{1}{xy^2zw^2} \\ 1 \otimes \frac{1}{y^2zw^2} \end{bmatrix} \leftarrow 1 \otimes \frac{1}{xy^2zw^2} \text{ in single summand} \right. \\ \left. \text{from } \mathcal{O}(0,-1) \otimes \mathbb{C}^2(\mathcal{O}(2,-4)) \right\}$$

$$\tilde{H}_0 \otimes \begin{cases} \text{still in} \\ \text{homology} \end{cases} \left\{ \begin{bmatrix} -1 \otimes \frac{1}{xy^2zw^2} \\ 1 \otimes \frac{1}{y^2zw^2} \end{bmatrix} \text{ in any summand? } \neq \text{no} \right.$$

$$\tilde{H}_1 \otimes \mathbb{C}^0$$

$$\begin{array}{ccc} F_0 & F_1 & F_2 \\ R(0,0,2,-3) & 2R(1,-1,2,-4) & R(0,-1,2,-4) \\ \oplus & \oplus & \\ R(1,-1,3,-4) & 2R(0,-1,3,-4) & \end{array} \left\{ \begin{bmatrix} -z \otimes 1 \\ x \otimes 1 \\ -1 \otimes z \\ 1 \otimes x \end{bmatrix} \right.$$

$$\left[\begin{array}{cc} w \otimes xy & w \otimes yz \\ -x \otimes 1 & -z \otimes 1 \end{array} \right] \text{ (from } M_2 \text{)}$$



4 edges $\rightarrow$ 4 vertices $\rightarrow$ 1 $\otimes$ 3

in each Čech complex

$$\left[\begin{array}{cc} w \otimes xy & w \otimes yz \\ -x \otimes 1 & -z \otimes 1 \end{array} \right] \left[\begin{bmatrix} -1 \otimes \frac{1}{xy^2zw^2} \\ 1 \otimes \frac{1}{y^2zw^2} \end{bmatrix} \right] = \left[\begin{array}{c} -w \otimes \frac{1}{y^2zw^2} + w \otimes \frac{1}{y^2zw^2} \\ -x \otimes \frac{1}{xy^2zw^2} + z \otimes \frac{1}{y^2zw^2} \end{array} \right]$$

$$\checkmark \emptyset \in \{\emptyset\}$$

$$\checkmark \emptyset \in \{\emptyset\}$$

$$\emptyset \in \{\emptyset\}$$

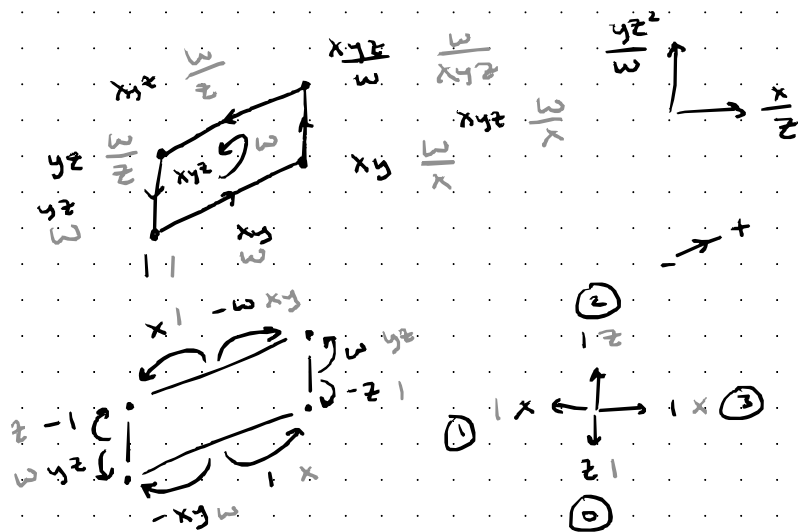
$$x \emptyset \in \{x\emptyset, \emptyset\}$$

$$x \emptyset \in \{x\emptyset, \emptyset\}$$

* also signs
in ss!

$$\{x\} \in 2\{x,z\} \quad \{x\} \in \{\{x\}, \emptyset\}$$

$$\{z\} \in 2\{x,z\} \quad \{z\} \in \{z\emptyset, \emptyset\}$$



b-tion 2 edges first, then top 2

for black first, then grey (doesn't matter)

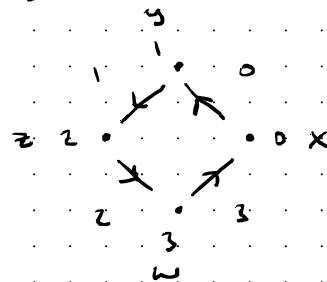
$$\begin{bmatrix} -xy \otimes w & yz \otimes w & -w \otimes xy & w \otimes yz \\ 1 \otimes x & -1 \otimes z & x \otimes 1 & -z \otimes 1 \end{bmatrix}$$

cure about
this part

$$\begin{bmatrix} z \otimes 1 \\ x \otimes 1 \\ 1 \otimes z \\ 1 \otimes x \end{bmatrix}$$

edges $\rightarrow$ pts $\rightarrow$ ~~z~~

$$\begin{bmatrix} -1 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \quad [1 \ 1 \ 1 \ 1]$$



$$\bullet z \quad [1] \quad \mathbb{C} \rightarrow \mathbb{C}$$

$$\left[0 \quad z \otimes \frac{1}{xy^2zw^2} \quad x \otimes \frac{1}{xy^2zw^2} \quad 1 \otimes \frac{1}{xy^2zw^2} \quad 1 \otimes \frac{1}{y^2zw^2} \right]$$

$$\uparrow \quad \left[0 \quad 1 \otimes \frac{1}{xy^2zw^2} \quad 0 \right]$$

$$\begin{matrix} \frac{1}{yzw} \\ \frac{1}{xzw} \\ \frac{1}{xyw} \\ \frac{1}{xyz} \end{matrix} \left[\begin{array}{ccc|ccc} & & & 0 & 1 \otimes \frac{1}{y^2zw^2} & \\ & & & 0 & 0 & \\ & & & 1 \otimes \frac{1}{xy^2zw^2} & 0 & \\ & & & 0 & 0 & \\ & & & 0 & 0 & \end{array} \right]$$

$$\begin{array}{ccc} 1 \times 2 & 1 \times 5 & 1 \times 3 \\ 4 \times 2 & 4 \times 5 & 4 \times 3 \\ 4 \times 2 & 4 \times 5 & 4 \times 3 \end{array}$$

$$\left[\begin{array}{ccc|ccc} w \otimes \frac{1}{yzw^2} & -z \otimes \frac{1}{y^2zw^2} & & & & \\ 0 & 0 & & & & \\ -w \otimes \frac{1}{yzw^2} & x \otimes \frac{1}{xy^2zw^2} & & & & \\ 0 & 0 & & & & \end{array} \right]$$

$$\begin{matrix} \bullet & \bullet \\ x & z \end{matrix}$$

$$\mathbb{C}^2 \rightarrow \mathbb{C}$$

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} !$$

* does this always happen?
what exactly

no need to associate homology of Δ to polytopes

have larger polytope where each point has subcomplex of Δ

$\mathbb{R}^n \times \Delta$ simplicial
complex
of fan
HKL
stratification

$n = \dim X$
 $d = \# \text{ of rays}$

cells of product = pairs of cells

→ double complex whose total complex

gives the homology of $\mathbb{R}^n \times \Delta$

* is the spectral sequence for the FM transform
a quotient of this?

pair of cell in $\overset{C}{\text{HKL strat}}$, cell in $\overset{\sigma}{\Delta}$
→ localization of S

↑ $[\pm 1]$ boundary map of Δ

$n-1$ -dim σ

← $S(-\text{label of } C)$
cellular
boundary $\times \sigma$

* need to take
specific deg to
remove dependence
on labels

$\dim C$

→ in some cases we are
taking unique maps
preserving the degree

or label cells in product
with the localizations

i.e. as submodules of Laurent polynomial ring all
maps are ± 1 (or 0)

* in a given fine degree, how do we get the complex of vector spaces topologically?

$$\text{fine degree } \frac{1}{y^2 w^2}$$

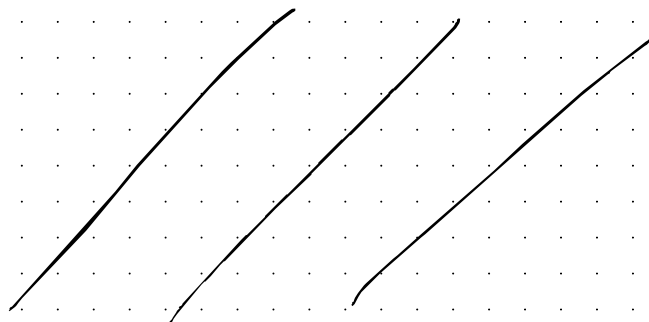
(origin in $\mathbb{R}^2, \{x, y\}$)

(origin, $\{z\}$)

$$\tilde{C}^1 \tilde{H}_1$$

$$\tilde{C}^1 \tilde{H}_0$$

$$\tilde{C}^0 \tilde{H}_1$$



F_0

F_1

F_2

(point in HHL, vertex of Δ)

$$S(x^0 y^0 z^0 w^0)_{xyzw} \oplus S(x^0 y^0 z^0 w^0)_{xyzw}$$

$$S(x^k | k \in M) \leftarrow S(x^k | k \in (0, 1, 1, 0) + M)$$

Inverted at 3 variables

* how does $\frac{1}{yuz}$ appear in infinitely many summands?

$\leadsto$ before quotient by L there are infinitely many cells

is but only finitely many inside polytope?

is there might also be strands with chains but no homology?

$\leadsto$ is this the same thing?

twist by, say, $\frac{xyz}{w}$

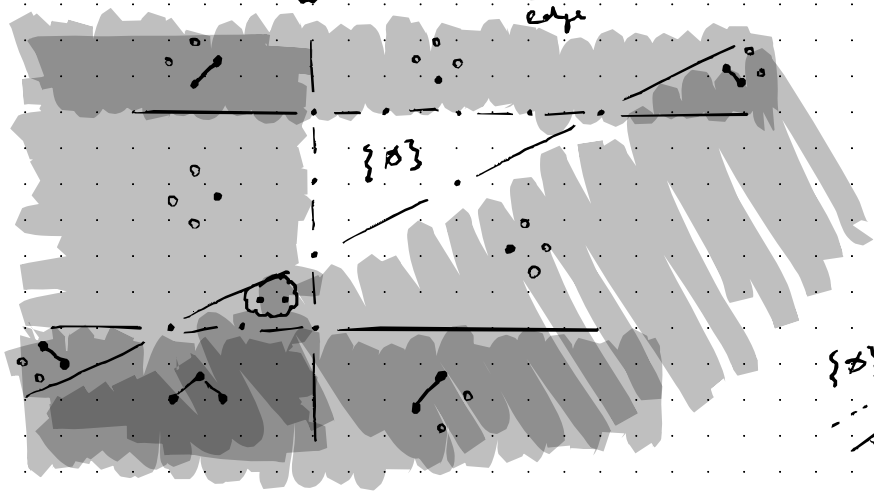
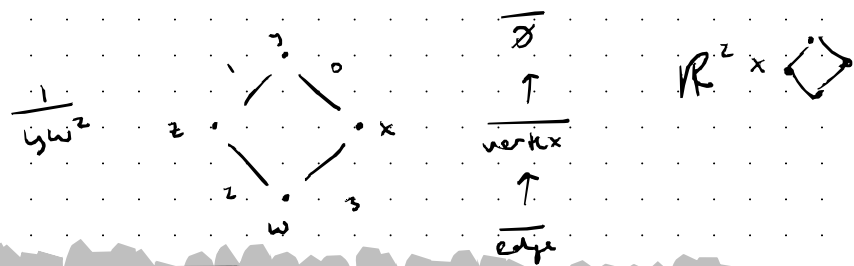
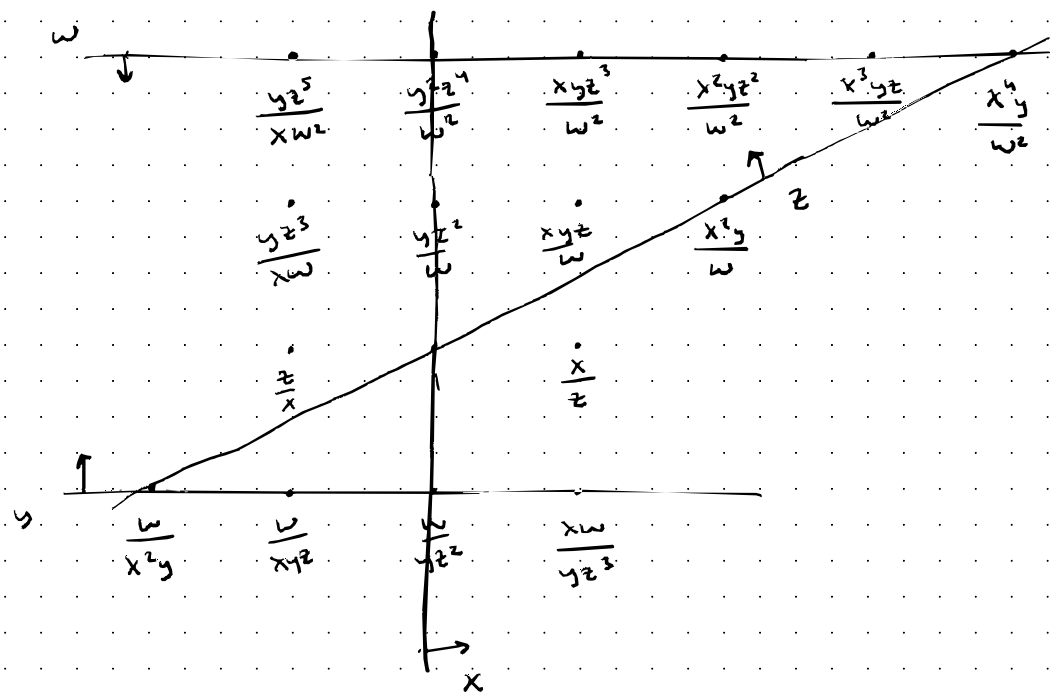
$$\begin{array}{c}
 S_{xyzw} \\
 \uparrow \\
 S_{yzw} \oplus S_{xzw} \oplus S_{xyw} \oplus S_{xyz} \\
 \uparrow \\
 S_{xy} \oplus S_{yz} \oplus S_{zw} \oplus S_{xz}
 \end{array}$$

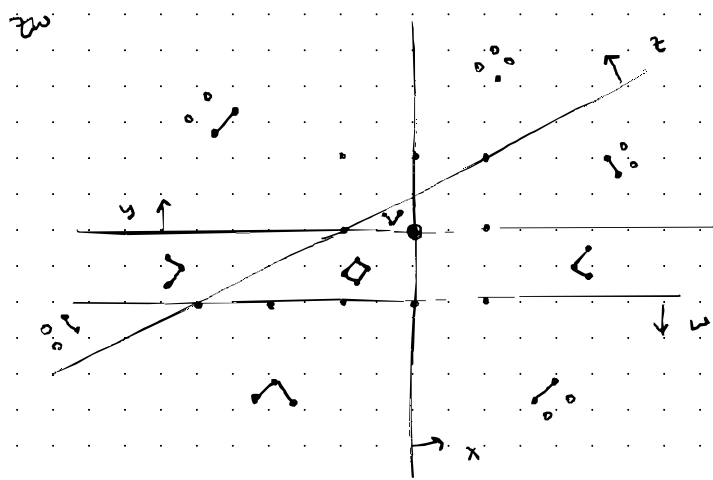
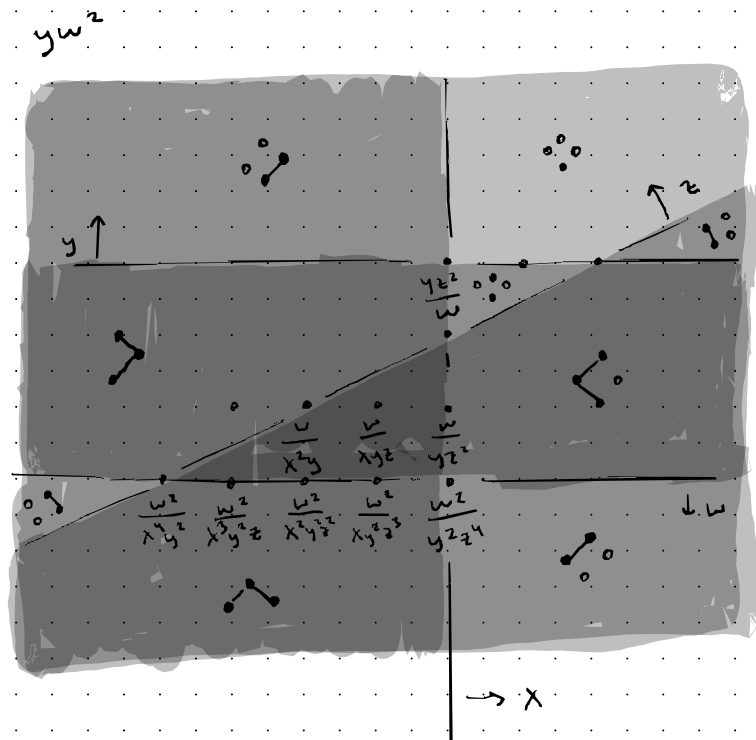
* in which summands can we find an elt to multiply

$$\frac{xyz}{w} \rightarrow \frac{1}{yuz}$$

i.e. in which summands does $\frac{1}{xyzw}$ exist?

$\leadsto$ only S_{xyzw}





* how do we account for the S_i in the
summands and not just the coefficients?

two different subcomplexes of Δ :

τ : X^{α_-} divides $X^{\hat{\tau}}_{\hat{\tau}}$ for α fine deg of
Cech complex

τ : $\tau \subseteq \sigma$ for σ subset of rays

α squarefree and nonpositive $\Rightarrow$

$\alpha \leftrightarrow$ subset α_- of rays

$$X^{\alpha_-} \mid X^{\hat{\tau}}_{\hat{\tau}} \Leftrightarrow X^{\alpha_-} \mid X^{\hat{\alpha}_-}_{\hat{\alpha}_-}$$

$$\Leftrightarrow \alpha_- \subseteq \hat{\tau} \Leftrightarrow \hat{\alpha}_- \geq \tau$$

$$\text{i.e. } \Delta_{\alpha} = \Delta_{\hat{\alpha}_-}$$

in MS:

local coh $\rightsquigarrow$ reduced cellular

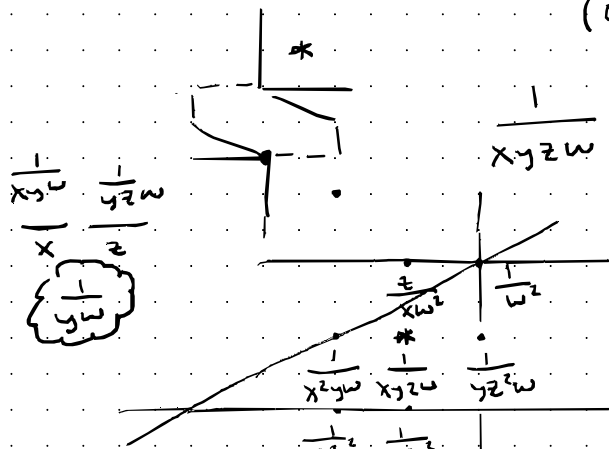
* use to skip steps?

in Sesh paper:

use this to compute local coh

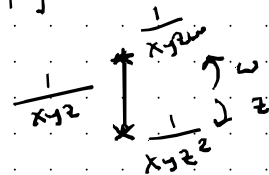
$\rightsquigarrow$ no need for $\mathbb{I}_{\Delta}$

$(0, -2)$



$$(1,0) + (-2,1) + (1,0) + (0,1) = (0,2)$$

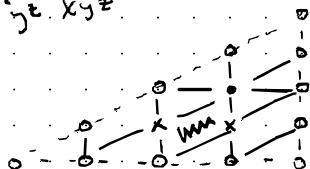
$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 2 \\ 0 & -1 \end{bmatrix}$$



$$\frac{1}{xy^2}$$

$(0, -1)$

$$\frac{x}{x^2 y^2} \quad \frac{x}{xy^2 z}$$



$$H^2 \quad S^3 \leftarrow S^4 \leftarrow S^{12}$$

H^1

H^0

$$\frac{1}{yw}$$

$S \otimes S$

$$S \otimes H^2(\mathcal{O}(0, -2))$$

$$\frac{1}{xyzw}$$

$$S(1, -1) \otimes S(1, -1)$$

$$S(0, 2) \otimes H^2$$

$$S(1, -1) \otimes H^2(\mathcal{O}(1, -3))$$

$$S(1, 1) \otimes H^2$$

$$F_0 \quad F_1 \quad F_2$$

$$S(0, -1) \otimes S(0, -1)$$

$$S(0, 1) \otimes H^2$$

$$S(0, -1) \otimes H^2(\mathcal{O}(0, -3))$$

2 1

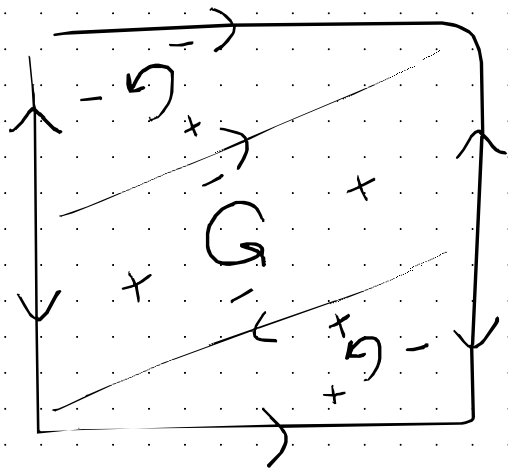
$$\begin{array}{r} 913 \\ 1812 \\ 530 \\ 711 \end{array} \quad \begin{array}{r} 610 \\ 4 \\ 2 \end{array}$$

$$\begin{bmatrix} w & 0 & 0 & w-x & z & y & 0 & 0 & 0 & x & y & 0 & 0 & 0 \\ -z & 0 & -x & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -z & 0 & -x & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

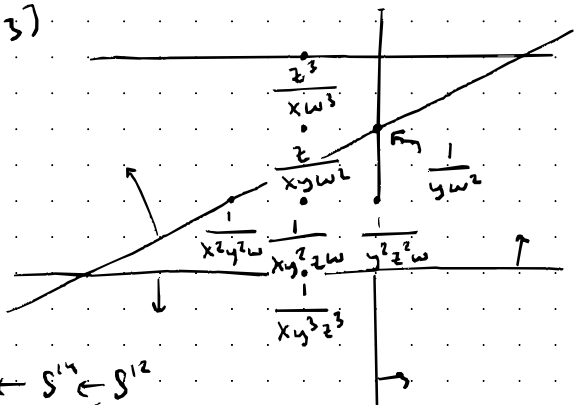
0	-x	0	0	1	0	0	0	0	0	0
0	0	-x	0	0	1	0	0	0	0	0
0	z	0	-1	0	0	0	0	0	0	0
0	0	z	0	0	-1	0	0	0	0	0
z	-w	0	0	0	0	y	0	0	0	0
x	0	-v	0	0	0	0	y	0	0	0
0	0	0	x	0	0	0	-1	0	0	0
0	0	0	0	x	0	0	0	-1	0	0
0	0	0	0	0	x	0	0	0	-1	0
0	0	0	0	0	0	x	0	0	0	-1
0	0	-z	0	0	0	1	0	0	0	0
0	0	0	-z	0	0	0	1	0	0	0
0	0	0	0	-z	0	0	0	1	0	0
0	0	0	0	0	-z	0	0	0	1	0

A 5x5 grid of dots with 20 asterisks scattered across it, representing a random distribution.

$$[y^2 x y \omega y z^2 x y z x^2 y z^2 x^2 y z x^3 y]$$



(2, -3)



$$S^3 \leftarrow S^{14} \leftarrow S^{12}$$

$$S^3 \leftarrow S^6 \leftarrow S^3$$

$$2x \quad 1x \quad 0$$

$$\begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 4 & 5 & 0 \\ \hline \end{array} \quad 2$$

$$\frac{z}{xyw^2} \quad \frac{1}{yw^2}$$

$$\frac{1}{xyw} \quad \frac{1}{xy^2zw}$$

$$Sec: \frac{1}{xy^2zw^2}$$

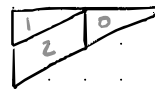
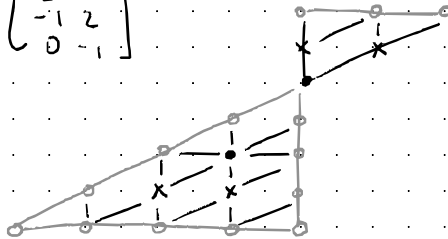
$$kwa: \frac{1}{yw}$$

$$\begin{array}{|c|c|c|c|} \hline * & * & * & * \\ \hline * & * & * & * \\ \hline \end{array}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & yz & xy \\ -z & 0 & -x & 0 & 0 & -1 \\ 0 & -z & 0 & -x & -1 & 0 \end{bmatrix}$$

$$\begin{array}{|c|c|} \hline * & * \\ \hline * & * \\ \hline * & * \\ \hline * & * \\ \hline \end{array}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 2 \\ 0 & -1 \end{bmatrix}$$



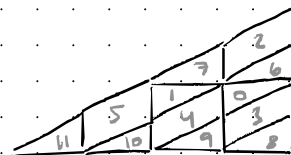
$$S^3 \leftarrow S^{14} \leftarrow S^{12}$$

$$S^3 \leftarrow S^6 \leftarrow S^3$$

$$\begin{bmatrix} A & 0 \end{bmatrix} \begin{bmatrix} B & 0 \\ E & C \end{bmatrix} \begin{bmatrix} 0 \\ D \end{bmatrix}$$

$$\begin{bmatrix} -x & 0 & 1 \\ 0 & -x & 0 \\ z & 0 & 0 \\ 0 & z & -1 \\ 0 & 0 & x \\ 0 & 0 & -z \end{bmatrix}$$

9	1	5	6
13	3	10	10
	8	4	4
	12	2	2
		7	11



•	• •	• •	•	• •
• •	•		• •	• • •

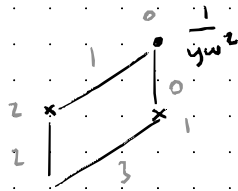
$$\begin{bmatrix} 1 & 0 & 0 & 0 & -x & z & yz & 0 & 0 & 0 & x & y & 0 & 0 & 0 \\ -z & 0 & -x & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -z & 0 & -x & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

[illegible]

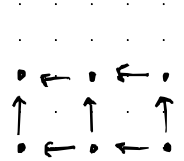
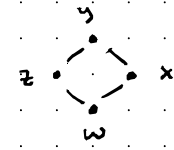
* same as previous!

* what is ϵ ? $S^3 \hookrightarrow S^{14}$

[* 122]



* now mapping other factor!

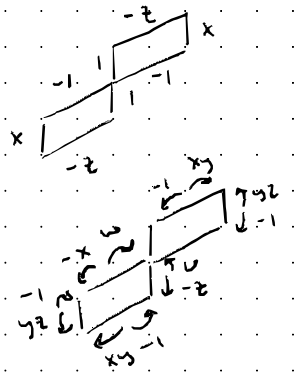


still in H^2

$$\begin{bmatrix} \frac{1}{y^2zw^2} \\ -\frac{1}{xy^2zw^2} \\ \frac{1}{xy^2zw^2} \\ -\frac{1}{xy^2zw^2} \end{bmatrix} \left[\frac{1}{xy^2zw^2} \otimes 1 \right]$$

$$[1, 1, 1, 1] S_{xyzw}$$

$$S_{zw} \oplus S_{xw} \oplus S_{xy} \oplus S_{yz}$$



$$\begin{bmatrix} \frac{1}{y^2zw^2} & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{xy^2zw^2} & 0 \\ 0 & 0 & \frac{1}{xy^2zw^2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{y^2zw^2} & 0 & -\frac{1}{y^2zw^2} & 0 \\ 0 & 0 & -\frac{1}{xy^2zw^2} & 0 \\ -\frac{1}{y^2zw^2} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{y^2zw^2} \otimes 1 \\ -\frac{1}{xy^2zw^2} \otimes 1 \\ \frac{1}{xy^2zw^2} \otimes x \\ \frac{1}{xy^2zw^2} \otimes z \end{bmatrix}$$

generator corresponding to

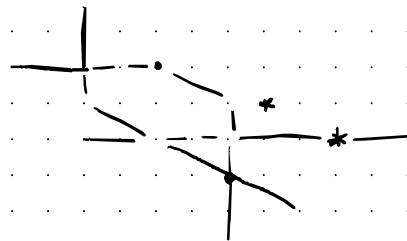
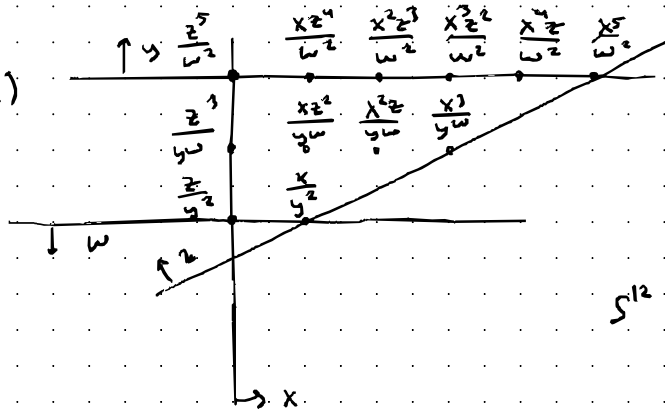
$$\square \text{ in } H^2(\theta(\frac{1}{xy^2zw^2}) \otimes \theta(\frac{1}{yw}))$$

generator of ?

$$H'(\theta(\frac{1}{yw})) \otimes \theta(\frac{1}{yw^2})$$

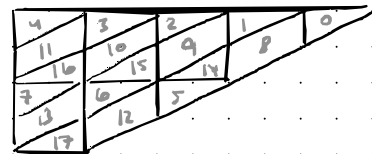
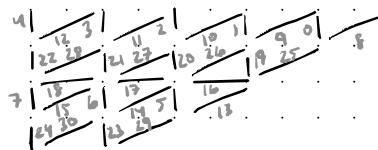
* times w !

(5, -2)

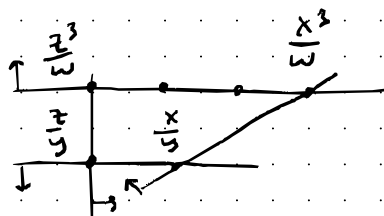


$$S^{12} \leftarrow S^{31} \leftarrow S^{18}$$

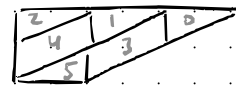
* 5
* 0
* 0
* 0
* 0



(3, -1)



* 2
* 1
* 0



$$S^2 \leftarrow S^{10} \leftarrow S^6$$

2 | 5 | 1 | 4 | 0 | 3
7 | 9 | 6 | 8 | . | .



$$\begin{bmatrix} -2 & 0 & 0 & -x & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -2 & 0 & 0 & -x & 0 & -1 & 0 & 0 & -1 \\ 0 & 0 & -2 & 0 & 0 & -x & 0 & -1 & 0 & 0 \end{bmatrix}$$

-x	0	0	1	0	0
0	-x	0	0	1	0
0	0	-x	0	0	0
z	0	0	0	0	0
0	z	0	-1	0	0
0	0	z	0	-1	0
0	0	0	x	0	-1
0	0	0	0	x	0
0	0	0	-z	0	0
0	0	0	0	-z	1

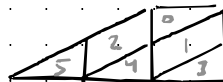
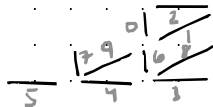
$(-3, 1)$

$$\frac{w}{x^2} \quad \frac{w}{x^2 z} \quad \frac{\frac{w}{x}}{x^2} \quad \frac{\frac{w}{z}}{z^2}$$

$$\begin{bmatrix} w & w & x & y & k_y \\ -z & -x & 0 & -1 & -1 \end{bmatrix}$$

→
←

$$\begin{bmatrix} -x & 1 & 0 \\ z & -1 & 0 \\ w & 0 & y \\ 0 & x & -1 \\ 0 & -z & 1 \end{bmatrix}$$

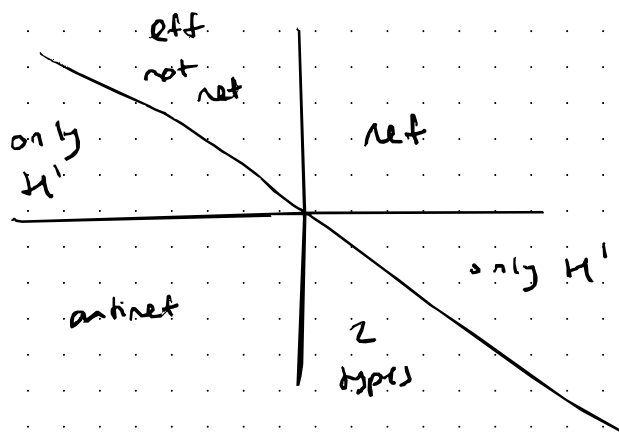


		上	上	上	上	
		中	中	中	中	
下	下					下

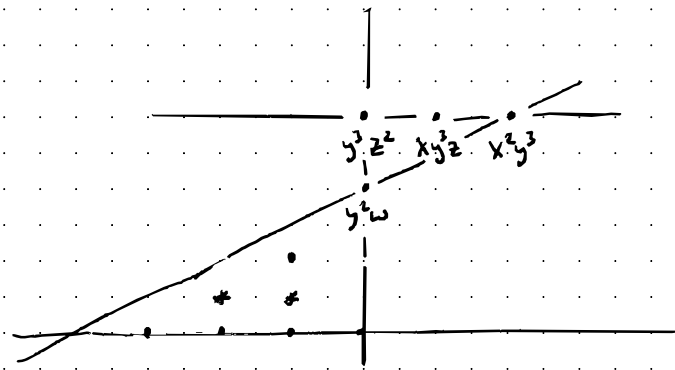
$$\begin{bmatrix} 1 & 0 & -x & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -x & z & 0 & yz & 0 & xy \\ 0 & 0 & 0 & 0 & -x & z & 0 & yz & xy \\ -z & -x & 0 & 0 & 0 & 0 & -1 & 0 & -1 \end{bmatrix}$$

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	不		不

-x	0	1	0	0	0
z	-1	0	0	0	0
-w	0	0	0	0	0
v	0	0	y	0	0
u	0	0	0	y	0
t	0	0	0	0	y
s	0	x	0	-1	0
r	0	0	x	0	-1
q	-z	0	1	0	0
p	0	-z	0	1	0



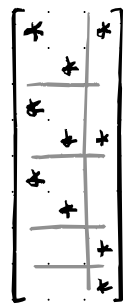
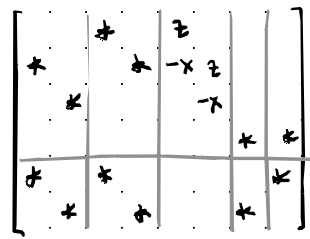
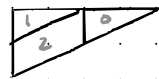
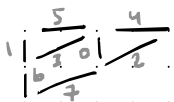
$(-4, 3)$



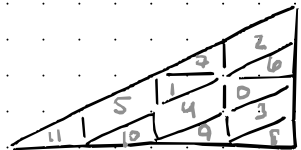
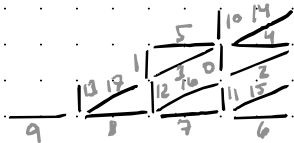
$$S^6 \leftarrow S^{10} \leftarrow S^{12}$$

$$S^6 \leftarrow S^8 \leftarrow S^3$$

$x^2 y^2$
 $x^2 y^2$
 0



$x^2 y^2$
 $x^2 y^2$
 0



$$\begin{bmatrix} w & w & * & * & * & * \\ -z & -x & 0 & -1 & -1 \end{bmatrix}$$

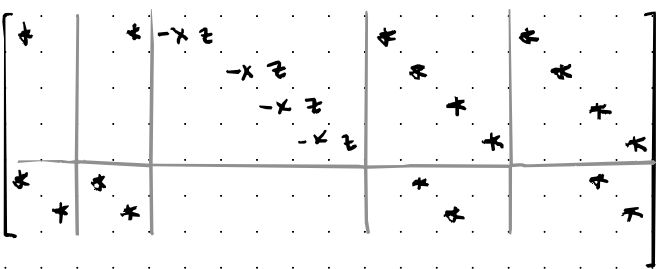
$$\begin{matrix} \leftarrow & \rightarrow \\ -x & z \end{matrix}$$

$$\begin{bmatrix} -x & 1 & 0 \\ z & -1 & 0 \\ w & 0 & y \\ 0 & x & -1 \\ 0 & -z & 1 \end{bmatrix}$$

$$\frac{y^2 z^2}{x} \quad y^2 w$$

$$\frac{y^2 w^2}{x} \quad \frac{y^2 w^2}{z}$$

$$\frac{y^2 w^2}{x^2} \quad \frac{y^2 w^2}{z^2}$$



$$\begin{bmatrix} x & x & x & x & x \\ 1 & 1 & 0 & -1 & -1 \\ -2 & -x & 2 & 2 & 2 \end{bmatrix}$$

$\uparrow$
 $\downarrow$
 x
 2

$$\begin{bmatrix} -x & 1 & 0 \\ 2 & -1 & 0 \\ 2 & 0 & 5 \\ 0 & x & -1 \\ 0 & -2 & 1 \end{bmatrix}$$

$\cdot$
 2
 0050
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